ECE556 Project: Part 3

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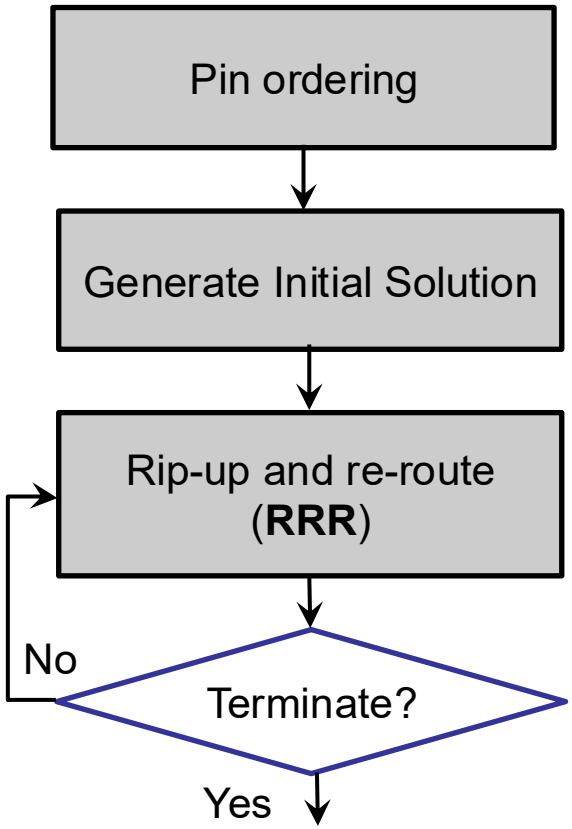
Routing Framework

- Components for project
  1. Pin ordering
  2. Generate initial solution and assign initial edge weights
  3. Rip-up and reroute
     - 3.1 "Net" ordering
     - 3.2 RRR each net based on min-cost routing; update edge weights at each step
  4. Termination condition



Termination Condition

- Min time: 5 min (ensures RRR is applied)
- Max time: 30 min
- Can be anything in between based on optimizing the quality metric below:

  Q = TOF X (1 + RUN/15)

  - where TOF is total overflow (sum of edge overflow) of your final routing solution <u>as reported by the evaluation script</u>
  - RUN is runtime of your executable <u>in minutes</u> on a free tux machine (best-tux.cae.wisc.edu)
  - Notice that 15 minutes is used as a reference for your runtime so any runtime beyond 15 minutes will result in scaling up TOF accordingly

Project Grading

- Project is 25% of your overall grade
- Part 1: 20% (due 3/12)
  - Full grade if passes evaluation script for the adaptec1 benchmark
- Part 2: 50% (due 4/14)
  - Based on meeting minimum requirements
- Part 3: 30% (due 4/28)
  - Based on value of Q (quality metric) of your submission
  - The quality metric Q will only be measured if the minimum requirements from part 2 are satisfied. If they are not satisfied, looking at the quality metric will be irrelevant, and no grade will be given for this portion.

Project Part 3

- Use any set of techniques that you desire to improve Q
  - Make the code faster
    - Faster maze routing? Improved A*? Parallelize existing code (should run on best-tux machines)?
  - Improve the quality, i.e., smaller TOF
    - For example, by improving the initial solution?
  - Better termination condition (stop time versus solution quality) to get a better Q
    - Stop quickly after generating a fairly-good solution?
  - Other ideas that you like to explore (such as improving pin ordering, etc.)

Project Part 3

- Steps for improving a component
  - Identify a component in your overall routing framework which can improve the evaluation metric
  - Think about a procedure to improve it, compared to how it is currently done, and think about why it should result in improvement before implementing it
  - Implement and evaluate by running some experiments to show how much your improvement has helped
    - Example: To evaluate the effectiveness of a new pin ordering algorithm: (1) create a table to measure for each design how the evaluation metric has improved using (1) the old pin ordering, and (2) your pin ordering

Project Part 3

- Each group member individually implements at least one improvement and prepares at most two-page report based on the following outline:
  - Explain the identified component
  - What is the opportunity for improving it? What is its current limitation that has potential room for improvement?
  - The procedure implemented by you for improving it
  - Evaluation on the provided benchmark to measure to what extent your improvement has helped
    - **Example:** If implementing a new pin ordering algorithm, create a table to measure for each design how the routing quality (in terms of different metrics that you identify as relevant) improve for (1) using the old pin ordering, and (2) using your pin ordering

What to Submit?

- Each group submits a single zipped file labeled "ece556-p3.zip" which should have a directory called “ece556-p3” to include the following files:
  - (1) ece556.cpp (2) ece556.h (3) main.cpp (4) Makefile (5) ReadMe
  - The zipped file may include additional source files but may not include object file (.o) or executables
  - Only include the best of all individual improvements (e.g., combined/integrated or some of the improvements), essentially the best version that you have as a group
  - Do not submit individual improvements
  - One submission per group

What to submit?

- Fill the following table in your ReadMe file and your final report
- Deadline for this is Tuesday April 28

Design	Your Part 2 submission (as a group)			Your Part 3 submission (best as a group)		
	RUN	TOF	Q	RUN	TOF	Q
Adaptec1						
Adaptec 2						
Adaptec 3						

What to submit?

- Each group member submits a separate report (at most 2 pages) using the guidelines in Slide 7.
- The report should be submitted by Thursday April 30 by each individual student.
- Include a table as below for the individual improvement:

Design	**Your Part 2 submission (as a group)**			**Your Part 3 (individual improvement)**		
	RUN	TOF	Q	RUN	TOF	Q
Adaptec1						
Adaptec 2						
Adaptec 3						

Results of Part 2

	-d=1 -n=1						
	adaptec 1		adaptec 2		adaptec 3		
Team Name	**TOF**	**RUN**	**TOF**	**RUN**	**TOF**	**RUN**	
The Floorplanners	**285,773**		29,112		**320,315**		**RANK**
FK Madison	369,044		34,982		530,107		**1**
Boolean Hooligans	310,319		**13,837**		218,716		**2**
JM	403,242		36,710		558,954		**3**
DK	**299,095**		29,171		357,572		
MM	394,272		83,402		558,570		
DP	423,808		139,674		1,131,600		
Business Majors	438,443		37,306		616,934		
PC	**153,139**		**716**		**86,208**		
MT	328,064		17,090		738,538		
AI	1,040,629		**9,621**		1,146,436		
MK	1,219,854		277,120		2,438,142		
OS	403,119		77,925		671,802		
IS	656,045		176,050		1,245,907		

- Only use as a point for comparison for part 3.
- Numbers are taken from the README file; not from the evaluation script.
- Part 3 will also be evaluated using some hidden designs.